Unit 2.2 Printables Teacher Key: Inherited Instructions

Standard: HS-LS3-2. Make and defend a claim based on evidence that inheritable genetic variations may result from (1) new genetic combinations through meiosis, (2) viable errors occurring during replication, and/or (3) mutations caused by environmental factors. Targets are labeled LS3-2.1 to LS3-2.5 (priority) and LS3-3.1, LS3-3.2 (supporting).

Data caveats

• All numbers and counts in the printables are illustrative, made up for class unless a source is cited on the item. They are not measurements from a study.

• Coin-flip, card, and Monster Genetics results vary by group. Sample answers show the method and the expected pattern, not one right draw.

• Codons and amino acids in the mutation printable follow the standard genetic code. The example message is made up.

• The class sample (22 green of 30 = 73%) is a made-up result. Real class tallies of 25 to 30 can differ from 75% by 10 or more points.

How to use this key

Items are in lesson order and match the Printables index. Sample answers are model answers, not the only acceptable wording: accept any answer that uses the evidence correctly. Notebook warm-ups and exit tickets are listed under each lesson.

Lesson 1: The Puppy Puzzle

Targets: LS3-2.1: I can name three ways new genetic variation arises.

Puppy litter data

Puppy A: black and floppy are like Mom; solid is like Dad. Puppy B: floppy and spotted are like Mom; brown is like Dad. Puppy C: black and spotted are like Mom; pointed is like Dad. Puppy D: floppy is like Mom; brown and solid are like Dad.

Every puppy has some traits like Mom and some like **Dad**. No two puppies have the same **combination** of traits (all four rows differ).

Notebook sample answers

Notice and wonder: answers vary (e.g., I notice every puppy has some traits like Mom and some like Dad; I wonder why no two are the same).

Three possible explanations: (1) information is mixed in different combinations, (2) information changes when it is copied, (3) the environment has an effect.

Key idea: Parents -> **information** -> Offspring -> similarities and **differences**.

Exit ticket: e.g., One explanation is that the information is combined in different ways. The evidence is that no two puppies have the same combination of traits.

Lesson 2: Mixing Information

Targets: LS3-2.1: I can name three ways new genetic variation arises.

Allele cards: Parent 1 and Parent 2

There is no single correct card draw. Any draw that takes exactly one card from each of the 6 piles is correct. The 6 piles: P1 fur, P1 ears, P1 pattern, P2 fur, P2 ears, P2 pattern. Each pile has 2 cards.

Card table: 3 offspring

Offspring 2 and 3 vary by draw. Check that each has one card from each of the 6 piles and that the two offspring are not identical. If a group draws two identical offspring, have them draw again: it is possible, but unlikely.

Sample sentence: Offspring 2 and 3 are different because each got a different combination of cards.

Notebook sample answers

Warm-up: Idea 1: information is **combined** in different ways. Idea 2: information **changes**. Idea 3: the **environment** has an effect.

Analyze the pattern: Every offspring got information from **both** parents. Each offspring got a **different** combination of cards. Different combinations can lead to different **traits**.

Revise our model: Offspring resemble their parents because they receive **information** from both. Offspring may differ because **the combination is different**.

Three sources of variation: 1. New **combinations** of alleles. 2. **Errors** when DNA is copied. 3. Mutations caused by the **environment**.

Lesson 3: Genetics Notes, Part 1: Genes and Alleles

Targets: LS3-3.1 (supporting): I can use a Punnett square to show the possible allele combinations from a cross.

Notebook sample answers

Warm-up: A **chromosome** is one long DNA molecule. A **gene** is a section of DNA that codes for one protein, and proteins help produce a **trait**.

Heredity: Genetics is the study of **heredity**. Heredity is the passing of **traits** from **parents** to **offspring**. The width of a tiger's stripes is a trait **inherited** from its parents.

Chromosomes: thread-like structures called **chromosomes**; they come in **pairs**; you usually have **two** copies of each, one from each parent.

Genes make proteins make traits: **DNA** is found within chromosomes. A **gene** is a short segment of DNA. Genes contain instructions that make **proteins**, which determine **traits**. A tiger's **DNA** produces **proteins** that determine the width of its **stripes**.

Alleles: individuals inherit **two** forms of each **gene**, one from each parent. The variations are **alleles**, written with lowercase and uppercase **letters**.

Dominant vs. recessive: Dominant: **always** expressed when present; **capital** letters. Recessive: expressed if **two** copies are present; **lowercase** letters.

Black or brown cats: A cat with Bb is **black** because **B** is dominant. A cat needs **two** copies of b to be brown.

Exit ticket: dominant; two.

Lesson 4: Genetics Notes, Part 2: Genotype, Phenotype, and Punnett Squares

Targets: LS3-3.1 (supporting): I can use a Punnett square to show the possible allele combinations from a cross.

Genotype to phenotype: pea plants

Genotype	Phenotype	Homozygous or heterozygous?
TT	Tall	Homozygous
Tt	Tall	Heterozygous
tt	Short	Homozygous

Tt is **tall** because T is dominant. tt is **short**.

Practice: TT; Tt; tt.

Punnett squares: steps and worked example

Tt × tt

	T	**t**
t	Tt	tt
t	Tt	tt

TT × Tt

	T	**T**
T	TT	TT
t	Tt	Tt

Tt × tt: 2 of 4 tall, 2 of 4 short. TT × Tt: 4 of 4 tall (2 TT, 2 Tt). Each box shows one possible **combination** of alleles.

Notebook sample answers

Warm-up: dominant = **capital**; recessive = **lowercase**.

Genotype and phenotype: the genotype is the **genetic** makeup (pair of **alleles**). The phenotype is the observable **traits**; for example **eye color**.

Homozygous and heterozygous: two of the **same** allele = **homozygous** (purebred); two **different** alleles = **heterozygous** (hybrid). Green: CC, dd, AA. Orange: Ee, Gg, Hh.

Practice: Cross 1 (Tt × tt): 2 of 4 tall and 2 short. Cross 2 (TT × Tt): 4 of 4 tall (the slide asks for how many are tall; accept 4).

Exit ticket: In Tt × Tt, **1** out of 4 boxes show a short plant (tt).

Lesson 5: Monster Genetics, Part 1: Flip a Coin, Build a Monster

Targets: LS3-3.1 (supporting): I can use a Punnett square to show the possible allele combinations from a cross.

Monster Genetics: the coin rule and Table 2

Trait	Male genotype	Male phenotype
Eyes	Ee	two small eyes
Skin color	Gg	green
Teeth	tt	flat
Feet	Ff	webbed
Horn color	hh	yellow
Claws	Kk	long
Tail	cc	straight
Ears	AA	round
Hair	Pp	hairy
Wings	ww	no wings
Spots	Ss	spotted
Body size	bb	small

Monster Genetics sheet: Table 1, the female monster

Results vary by flips. Check each row: Heads+Heads = capital capital (e.g., EE); Heads+Tails or Tails+Heads = capital + lowercase, written capital first (Ee); Tails+Tails = lowercase lowercase (ee). Dominant phenotype shows if there is a capital letter.

Prediction check: for Eyes, a female Ee × male Ee gives a 25% chance of ee (one large eye); a female EE gives 0%; a female ee × male Ee gives 50%. This is done in Lesson 6.

Notebook sample answers

Warm-up: Each sex cell gets **one** allele. A coin flip can choose between B and **b**.

Compare: Our monsters are different because the flips gave different **combinations** of alleles.

Exit ticket: Each coin flip stood for the **allele** a sex cell passes on. Two flips gave the monster's **genotype**.

Lesson 6: Monster Genetics, Part 2: Predict the Baby

Targets: LS3-2.2: I can describe how making sex cells and fertilization mix alleles into new combinations.; LS3-3.2 (supporting): I can calculate the chance of each trait and compare it with real counts.

Punnett square: skin color and eyes

Gg × Gg

	G	**g**
G	GG	Gg
g	Gg	gg

The letters on the top and side are the **alleles** in each parent's sex cells. Gg × Gg: 75% green, 25% blue.

Student squares depend on their coin flips. Check: each box = 25%; percent dominant = (boxes with a capital letter) × 25. Male genotypes for the six traits: Eyes Ee, Skin color Gg, Teeth tt, Feet Ff, Horn color hh, Claws Kk. Example: female Gg × male Gg for skin color: 3 dominant boxes = 75%. Female Tt × male tt for teeth: 2 of 4 dominant = 50%.

Quick key if the female is Ee × Ee (male Ee): 3 of 4 dominant (75%), 25% one large eye.

Prediction vs. class count

The Punnett square predicted **75**% green. The sample class got about **73**% green. The numbers are close but not exact because every flip is **random**.

Class tally will vary. Expect values near 75% green for larger classes; small samples can differ by 10 or more points. This is the point: probability predicts a pattern, not each baby.

Notebook sample answers

Warm-up: A Gg parent makes sex cells with **G** or **g**.

Sex cells and fertilization: Meiosis makes **sex cells** with one allele of each gene. Crossing over happens when paired chromosomes **swap** pieces of DNA. Fertilization joins two sex cells to make a new **combination** of alleles.

Percent chance: In a Gg × Gg cross, **75**% of the boxes are green and **25**% are blue.

Class data: predicted **75**% green; class got about **73**%; close but not exact because every flip is **random**.

Lesson 7: When Copying Goes Wrong

Targets: LS3-2.3: I can explain how copying errors and things like UV light change genes, and when that change is inherited.

Four kinds of mutations

Blank	Answer
1	amino acid
2	effect
3	different
4	stop
5	terminates
6	shifted

Codon check: GAA and GAG both code for glutamic acid (Glu); GUA codes for valine (Val); UGG codes for tryptophan (Trp); UGA is a stop codon. Frameshift: deleting the first G in GAA shifts the reading frame (AUG AAC ACU GG...).

UV light and mutations

With more UV light, there are **more** mutations. The 2 mutations with no UV came from **copying** errors. Points: (0, 2), (1, 6), (2, 11), (4, 23): the line rises, so more UV goes with more mutations.

About 21 of the 23 mutations at 4 hours (23 minus the 2 baseline) are above the no-UV level, so UV likely caused most of them.

Who passes it on? Five cases

Case	Inherit?	Why
A	No	Body (skin) cell
B	Yes	The cell makes sperm, so the change can be in the sperm that fertilizes an egg
C	No	Skin is a body cell
D	Yes	The change is in the egg, so it can be passed to offspring
E	No	Liver is a body cell

A change in a **body** cell is not passed on. A change in a cell that makes **eggs or sperm** can be.

Sample explanation: UV light damages DNA, which can change a gene. The change is inherited only if it happens in a cell that makes eggs or sperm.

Notebook sample answers

Warm-up: A mutation is a **change** in the DNA sequence. A mutation can change an **amino acid** in a protein.

How do mutations happen: A **copying** error that is not fixed becomes a mutation. UV light and some **chemicals** can also damage DNA.

Explain it: UV light can change a gene by **damaging** its DNA. The change is inherited only if it happens in a cell that makes **eggs or sperm**.

Lesson 8: Where Did This Variation Come From?

Targets: LS3-2.4: I can read data and make a claim, backed by that data, about what caused a genetic variation.

Case data: three cases

Case 1: **New combination.** Both parents are Bb, so no new allele appeared; the brown puppy is bb because it got a b from each parent.

Case 2: **Mutation (copying error).** The puppy's CCT allele is new because neither parent has it. The most likely source is a copying error (no extra exposure is described), not a new combination.

Case 3: **Environmental mutation.** High dose had 25 mutations per 1,000 sperm cells, no X-rays had 3. The extra mutations came from the X-rays. If copying errors alone caused every mutation, the groups would have similar numbers.

CER model for Case 3: Claim: The extra mutations came from an environmental mutation (the X-rays). Evidence: The high-dose group had 25 mutations per 1,000 sperm cells, but the group with no X-rays had 3. Reasoning: If copying errors alone caused every mutation, the groups would have similar numbers.

Notebook sample answers

Clues: New combination: every allele in the offspring was **already** in a parent. Copying error: a **new** allele that neither parent has, and no extra exposure. Environmental mutation: **more** mutations in the group that was exposed.

Case 2: The puppy's CCT allele is **new** because neither parent has it. The most likely source is a **mutation**, not a new combination.

Case 3: The high-dose group had **25** mutations. The group with no X-rays had **3**. The extra mutations came from **the X-rays**.

CER: Claim: The extra mutations came from **an environmental mutation**. Evidence: The high-dose group had **25** mutations per 1,000 sperm cells, but the group with no X-rays had **3**. Reasoning: If copying errors alone caused every mutation, the groups would have **similar** numbers.

Lesson 9: Defend Your Claim and Back to the Puppies

Targets: LS3-2.5: I can defend my claim against another explanation using the evidence.

The new coat pattern: two kennels

Evidence: Kennel A had **14** new patterns per 1,000 puppies. Kennel B had **3**. Defense: Copying errors happen in **both** kennels, so they cannot explain the **11** extra patterns.

Sample defense: I claim a chemical in Kennel A's water caused the new patterns. Kennel A had 14 per 1,000 and Kennel B had 3. Copying errors happen in both kennels, so they only explain about 3 patterns; they cannot explain the 11 extra in Kennel A.

Notebook sample answers

How to defend a claim: A counterclaim is **another** explanation. To defend my claim I show why the **counterclaim** does not fit the data.

The new coat pattern: see the printable key.

Write It: Claim: Puppies from the same parents are alike but not identical because of **new combinations of alleles**. Evidence: In the card activity, each offspring got a different **combination** of alleles. Reasoning: Sex cells and fertilization make new combinations. Copying errors and **environmental mutations** can also change DNA, and a change is inherited only if it is in **eggs or sperm**.

Lesson 10: Review Day: Prepare for the CSA

Targets: LS3-2.1: I can name three ways new genetic variation arises.; LS3-2.2: I can describe how making sex cells and fertilization mix alleles into new combinations.; LS3-2.3: I can explain how copying errors and things like UV light change genes, and when that change is inherited.; LS3-2.4: I can read data and make a claim, backed by that data, about what caused a genetic variation.; LS3-2.5: I can defend my claim against another explanation using the evidence.

My target tracker

Student responses vary. Look for: lowest target chosen first, a specific plan (review in Canvas, flashcards, redo the case data), and honest self-ratings.

Hydrangea soil data

The plants have the same **DNA**. Flower color varied because the **soil** changed how the gene works.

Not an inherited mutation: the plants are clones with the same DNA, and color changes with soil, so the environment affected how the trait was expressed. A mutation would change the DNA sequence and stay the same in any soil.

Notebook sample answers

Student responses vary: look for honest CFA scores, completed reviews, and a specific plan.

Lesson 11: CSA

Targets: LS3-2.1: I can name three ways new genetic variation arises.; LS3-2.2: I can describe how making sex cells and fertilization mix alleles into new combinations.; LS3-2.3: I can explain how copying errors and things like UV light change genes, and when that change is inherited.; LS3-2.4: I can read data and make a claim, backed by that data, about what caused a genetic variation.; LS3-2.5: I can defend my claim against another explanation using the evidence.

Notebook sample answers

Reflection: student responses vary. Look for the three sources of variation (new combinations, copying errors, environmental mutations) and that only changes in eggs or sperm are inherited.

Matching table: data to source of variation

Use this table to check Lesson 8 and Lesson 9 answers. Match each data pattern to its likely source of variation.

Data pattern	Likely source of variation	Why
Both parents are Bb; brown puppy is bb	New combination of alleles	The puppy has only alleles the parents already had.
Puppy has an allele (CCT) that neither parent has; no extra exposure	Copying error	A new allele with no exposure points to an error during copying.
25 mutations per 1,000 sperm cells after high X-ray dose vs. 3 with no X-rays	Environmental mutation	The exposed group had far more mutations than the unexposed group.
Kennel A: 14 new patterns; Kennel B: 3	Environmental mutation (chemical in water)	Copying errors happen in both kennels; the extra 11 need another cause.
Same plant DNA, flower color varies with soil	Environment affecting gene expression (not a mutation)	DNA is the same; the trait changed with soil.